Mathematics GCE O Level 2019 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Cameroon General Certificate of Education Board, Ordinary Level, June 2019, 0570 Mathematics 1. Multiple choice question paper, one and a half hours. Answer ALL 50 questions; all questions carry equal marks. Each question has four suggested answers A, B, C and D: choose one. Mark only one answer for each question. Calculators are allowed.

Multiple choice questions - choose the one correct answer, A, B, C or D.

1. When $\frac{1}{2}$ of $6 - \left(\frac{3}{2} \times 4\right)$ is evaluated, the result is

a) 3
b) 6
c) 0
d) -3

2. The value of 8 in 24.86 is

a) 8
b) $\frac{8}{10}$
c) $\frac{8}{100}$
d) $\frac{9}{10}$

3. Given that $-5 < 2$, the symbol "<" means

a) Greater than
b) Greater than or equal to
c) Less than
d) Less than or equal to

4. Simplifying $\frac{2}{7} + \frac{2}{5}$ gives

a) $\frac{4}{35}$
b) $\frac{12}{35}$
c) $\frac{4}{12}$
d) $\frac{24}{35}$

5. The number 163.35×10^{-3} expressed in standard form is

a) 1.6335×10^{-1}
b) 16335×10^{-2}
c) 1.6335×10^{1}
d) 16.335×10^{-2}

6. Expressing $204{,}865$ to three significant figures gives

a) $205{,}000$
b) $204{,}800$
c) $204{,}900$
d) $200{,}900$

7. A student left home and took 45 minutes to arrive at school at 8:00 am. The time the student left home is

a) 7:45 am
b) 6:15 am
c) 7:15 am
d) 8:45 am

8. A woman paid 120,00 FCFA for 6 kg of fish. The cost of $\frac{1}{2}$ kg of the fish is

a) 3,000 FCFA
b) 1,000 FCFA
c) 6,000 FCFA
d) 2,000 FCFA

9. On the map of Cameroon, the distance between Bali and Batibo is 11 cm. Given that the scale of the map is 1:200,000, the actual distance between Bali and Batibo, in km, is

a) 2 km
b) 2.2 km
c) 220 km
d) 22 km

10. Given the sets: $P = \{1, 2, 3, 4, 5\}$ and $Q = \{2, 5\}$. The relationship between the sets P and Q is

a) $P \cup Q = Q$
b) $Q \subset P$
c) $P \cap Q = \emptyset$
d) $P \subset Q$

11. The number of subsets of the set $T = \{a, b, c\}$ is

a) 3
b) 6
c) 8
d) 9

12. Given the statements

p: Ben studies hard

q: Ben will pass his exams

The statement *"If Ben studies hard then he will pass his exams"* is represented by

a) $p \Rightarrow q$
b) $p \vee q$

c) $
eg p \wedge q$
d) $p \wedge q$

13. Given the truth table in figure 1, Y represents

p	q	Y
T	T	T
T	F	F
F	T	F
F	F	F

Figure 1

a) $p \vee q$
b) $
eg p \wedge q$
c) $p \wedge q$
d) $p \Rightarrow q$

14. Given the function $f : x \mapsto 2x - 1$, the value of $f(-3)$ is

a) 5
b) 7
c) -5
d) -7

15. The inverse of the function $f(x) = 3x$ is

Options as printed: A) $\frac{x}{2}$ B) $\frac{3}{x}$ C) $\frac{3x}{2}$ D) $\frac{2}{3x}$

16. Given the mapping in figure 2, the range set is

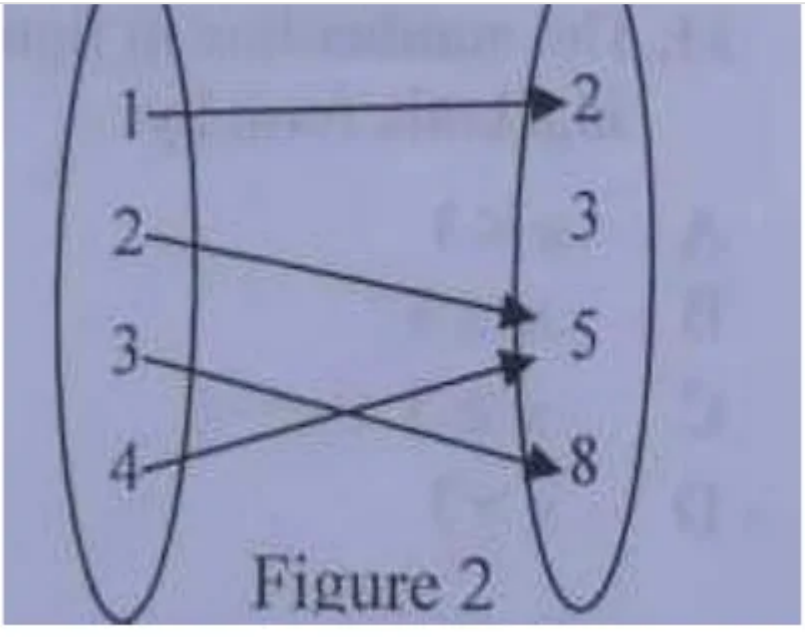
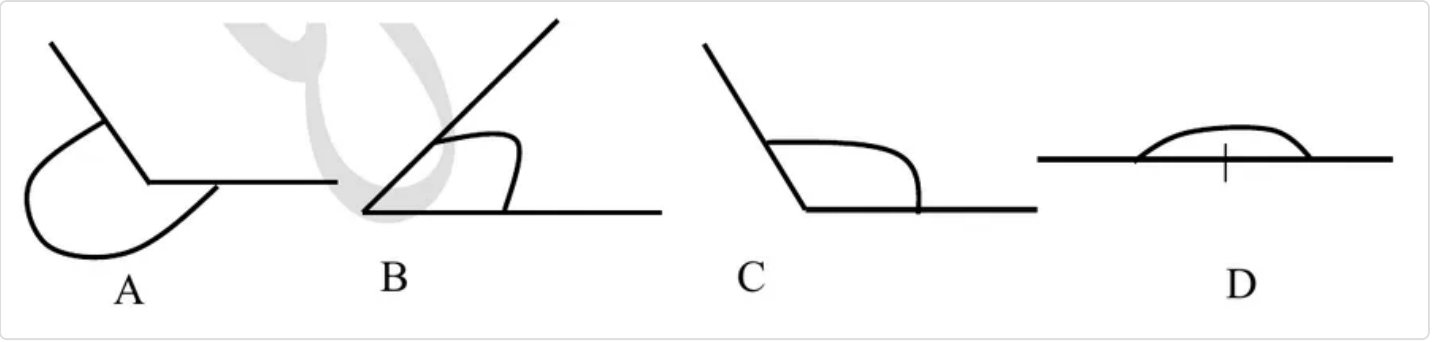


Figure 2

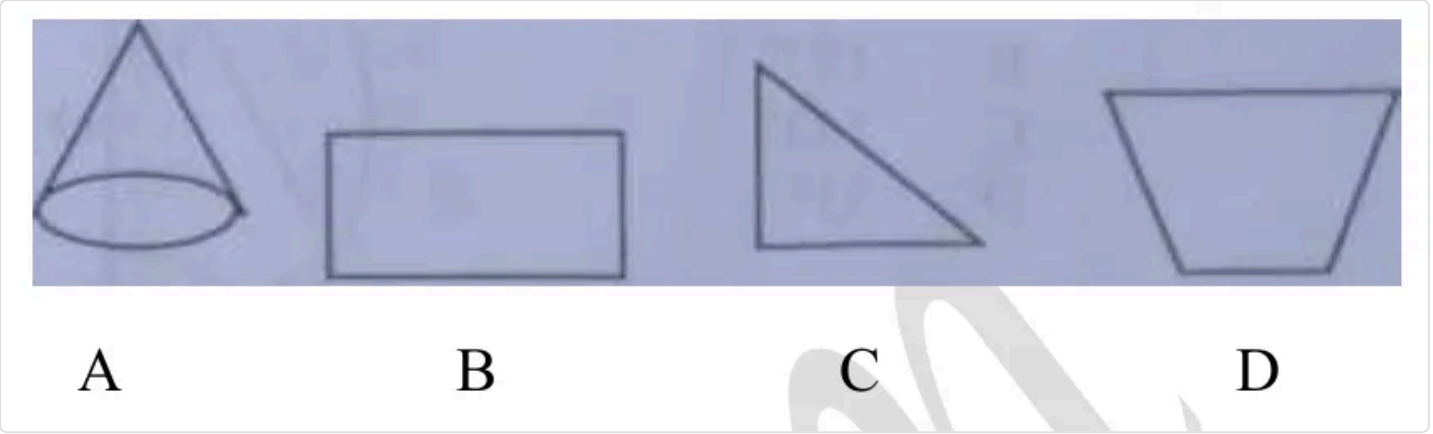
a) $\{2, 3, 5\}$
b) $\{2, 5, 8\}$
c) $\{1, 2, 3, 4\}$
d) $\{1, 2, 5, 5, 8\}$

17. The diagram that represents an acute angle is



a) A
b) B
c) C
d) D

18. The figure that represents a solid is



a) A
b) B
c) C
d) D

19. The sum of the exterior angles of any polygon is

a) 90°
b) 180°
c) 360°
d) 270°

20. The diagrams in figure 3 show two similar triangles. The area, P, of the larger one is

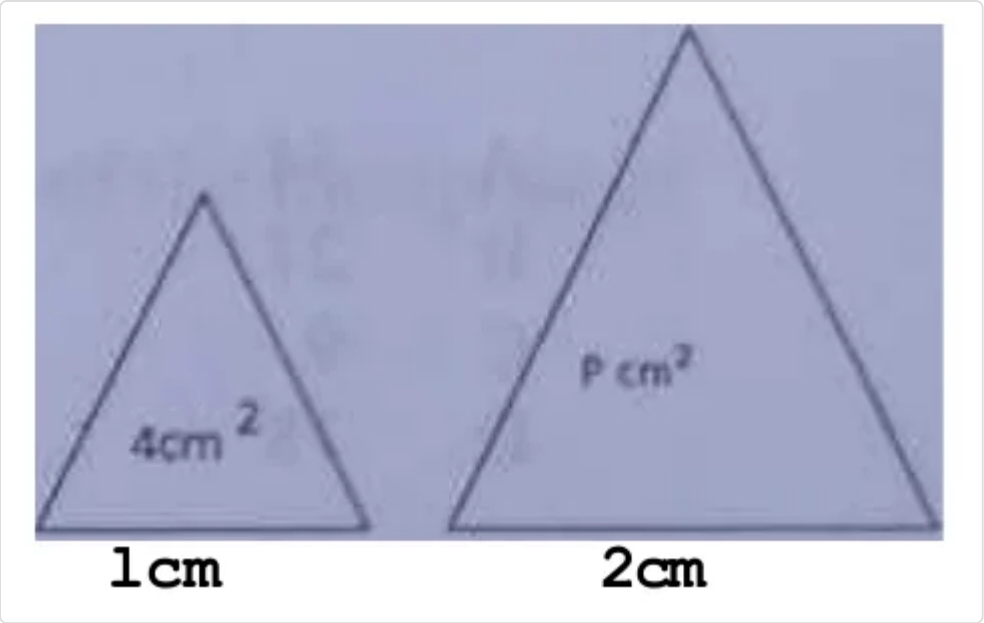
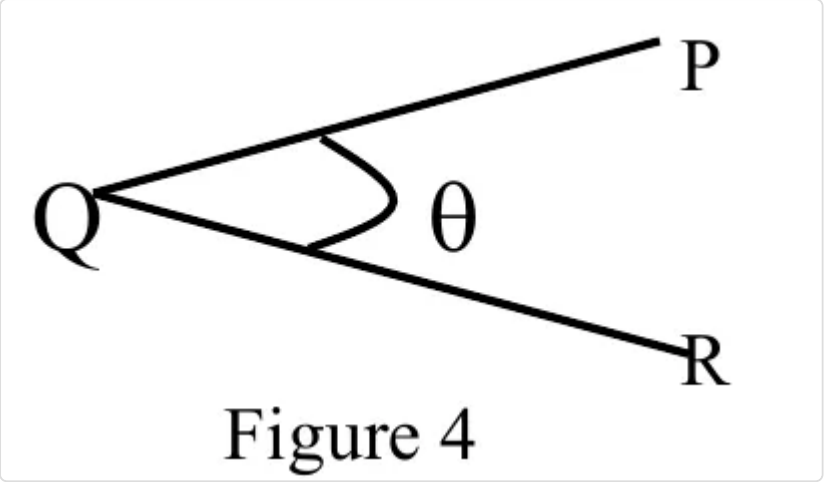


Figure 3

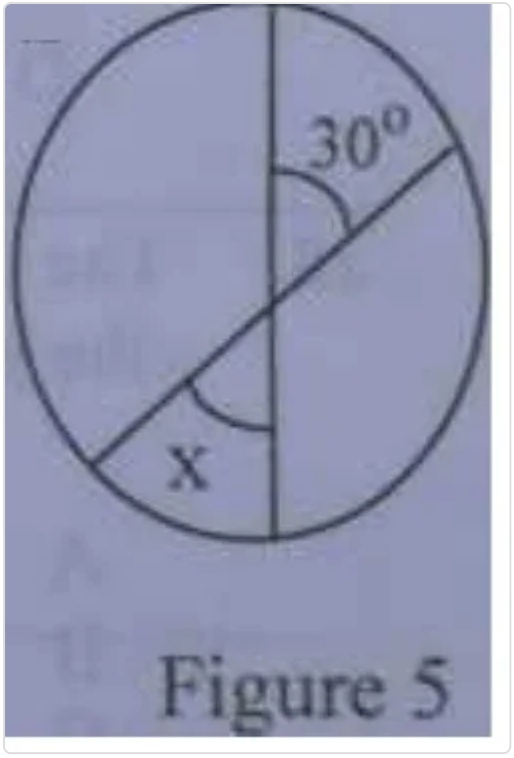
a) 16
b) 12
c) 24
d) 72

21. In figure 4, the arms of the angle θ are



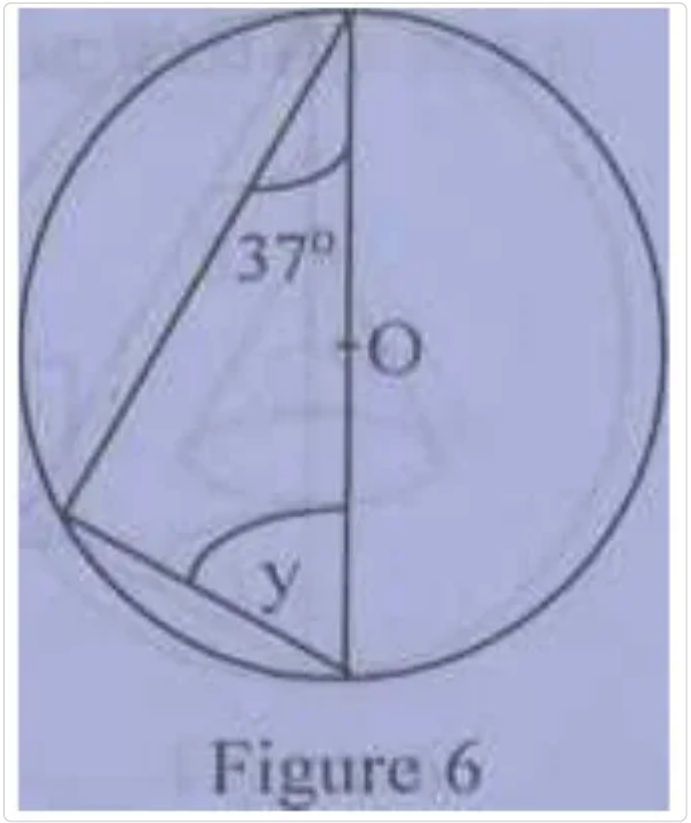
a) P and Q
b) PQ and PR
c) QP and QR
d) PQR and RQP

22. In figure 5, the value of angle x is



a) 150°
b) 60°
c) 45°
d) 30°

23. In figure 6, O is the centre of the circle. The value of y is

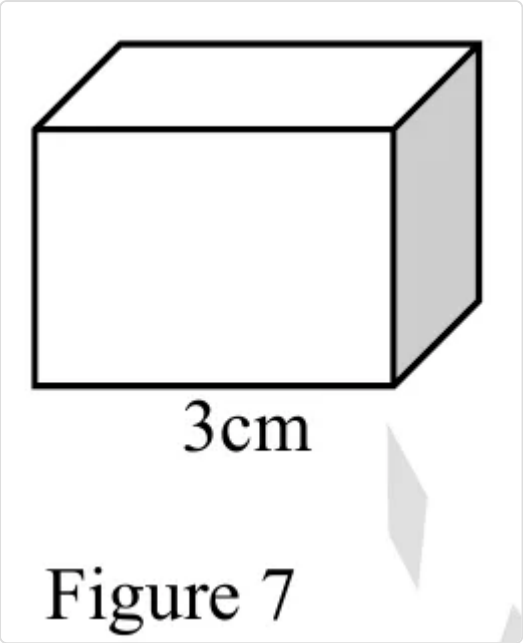


a) 37°
b) 127°
c) 143°
d) 53°

24. The diameter of a circle is 28 cm. The circumference, in cm, of the circle is

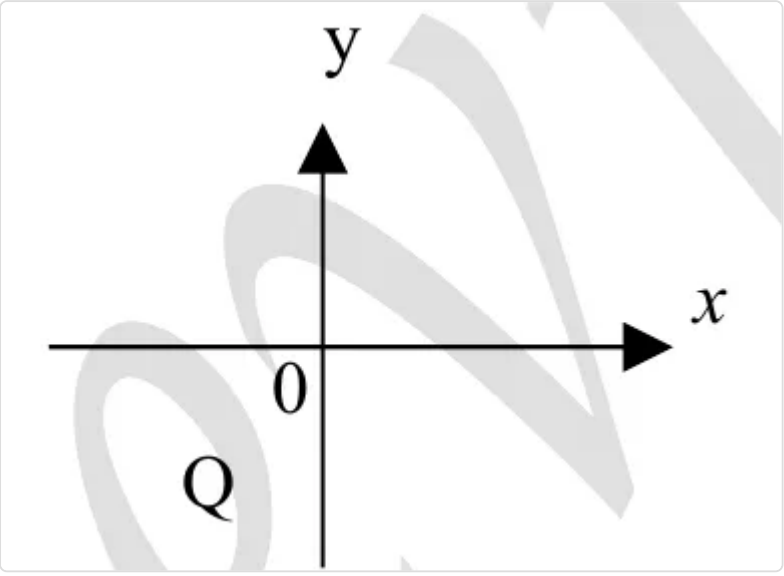
a) 7
b) 88
c) 14
d) 56

25. Figure 7 is a cube of side 3 cm. The total surface area in cm^2 is:



a) 54
b) 27
c) 9
d) 18

26. Figure 8 represents a Cartesian plane. The quadrant in which Q is found is

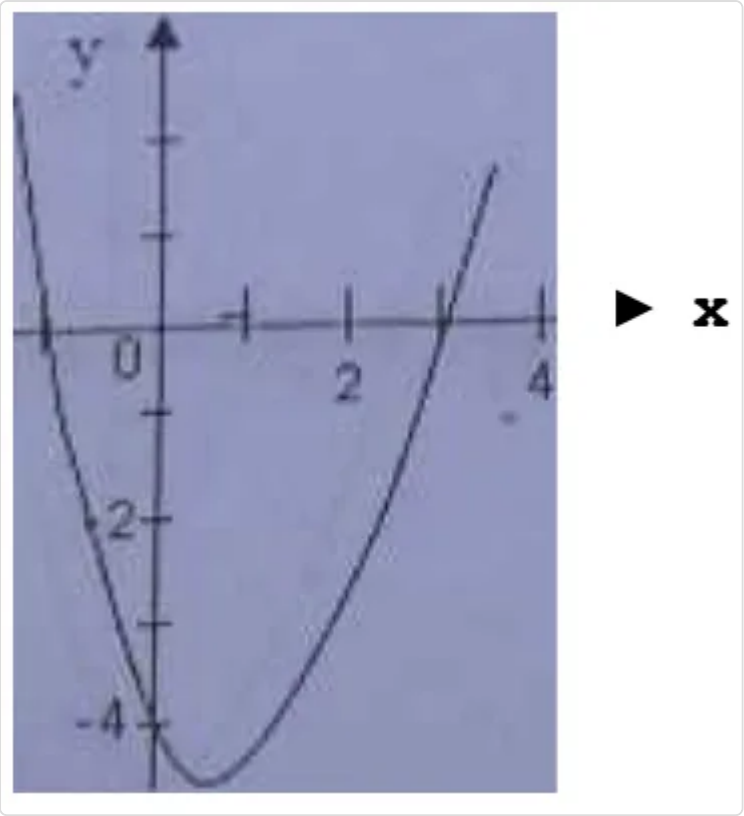


a) 1st quadrant
b) 4th quadrant
c) 2nd quadrant
d) 3rd quadrant

27. The gradient of the line $2y - 4x = 3$ is

a) -2
b) 4
c) 2
d) -4

28. The coordinates of the point where the curve cuts the y-axis, in figure 9, are given by



a) $(3, 0)$
b) $(0, -4)$
c) $(-1, 0)$
d) $(2, -2)$

29. In the expression $3x^2 - 15x$, the common factor is

a) $3x$
b) 5
c) $3x^2$
d) $5x$

30. Given the expression $8x + 2 - x + 8y$, the like terms are

a) $8x$ and $8y$
b) $8x$ and $-x$
c) 2 and $-x$
d) -8 and $8y$

31. Given that $y = -2$ and $x = 1$, then the value of $y^2 + x + 1$ is

a) 6
b) 4

c) -2
d) 0

32. Given that $ab^2 = xy$, then b expressed in terms of a, x and y is

a) $\sqrt{xy - a}$
b) $\sqrt{\frac{xy}{a}}$
c) $\sqrt{\frac{a}{xy}}$
d) $\sqrt{a - xy}$

33. The number line in figure 10 can be represented in algebraic form by

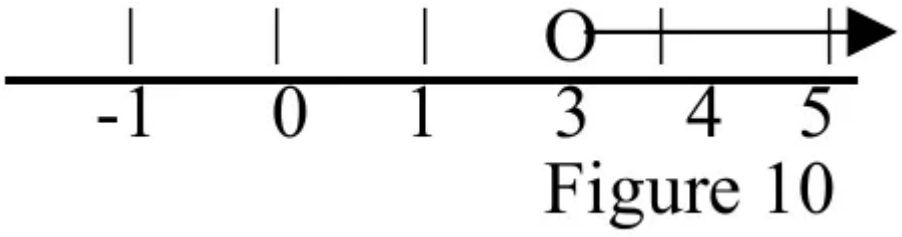


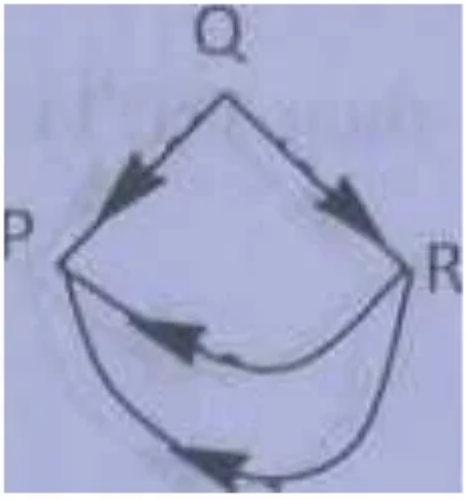
Figure 10

a) $x < 3$
b) $x \leq 3$
c) $x \geq 3$
d) $x > 3$

34. The solution of the linear inequality $2x - 3 < 5$ is

a) $x > 4$
b) $x \leq 4$
c) $x < 4$
d) $x \geq 4$

35. Figure 11 represents a network. The number of vertices are



a) 1
b) 2

c) 3
d) 4

36. The value of x in the equation $2^{2x} = 8$ is

a) $\frac{2}{3}$
b) 2
c) 4
d) $\frac{3}{2}$

37. In figure 12, PQR is a right angled triangle, the length of the side PR in cm is.

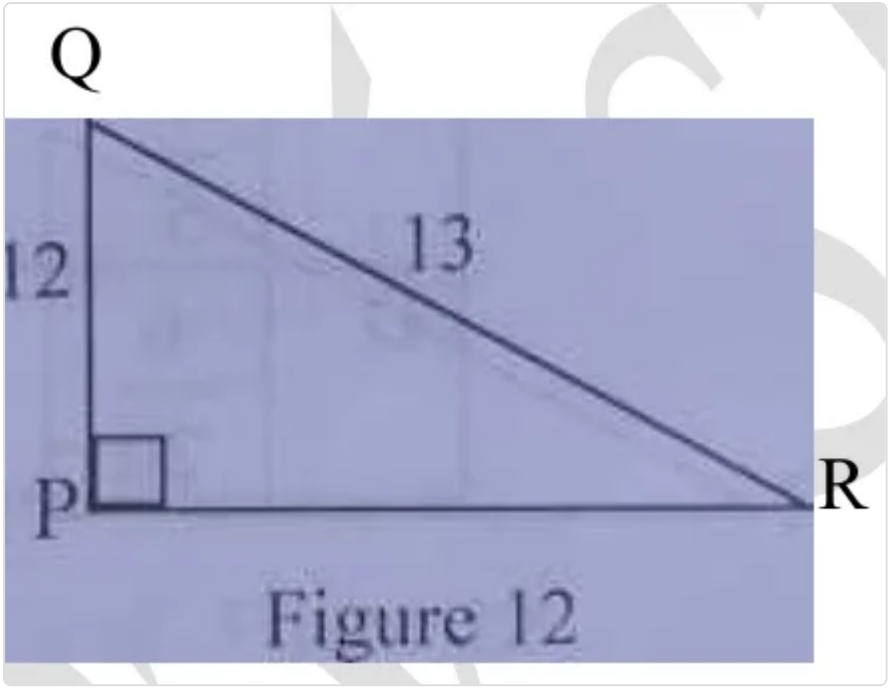


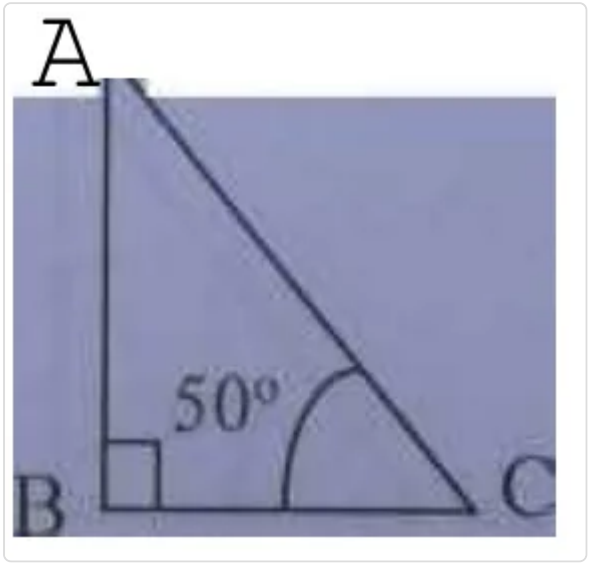
Figure 12

Options as printed: A) 25 B) -5 C) 1 D) .....

38. Given that $\sin\theta = \frac{3}{5}$, then $\tan\theta$ is

Options as printed: A) $\frac{4}{5}$, B) $\frac{4}{5}$, C) $\frac{4}{5}$, D) $\frac{4}{5}$,

39. Figure 13 shows the angle of elevation of A from C as 50°. The angle of depression of G from A is



a) 140°
b) 50°
c) 90°
d) 40°

40. Given that $\overrightarrow{OP} = 3\mathbf{i} - 5\mathbf{j}$ and $\overrightarrow{OQ} = -2\mathbf{i} + \mathbf{j}$, the vector $\overrightarrow{QP}$ is

a) $5\mathbf{i} - 6\mathbf{j}$
b) $5\mathbf{i} - 4\mathbf{j}$
c) $\mathbf{i} - 4\mathbf{j}$
d) $-5\mathbf{i} + 6\mathbf{j}$

41. Given the vector $\overrightarrow{OP} = \mathbf{i} + \mathbf{j}$, the direction of this vector is

a) 30°
b) 60°
c) 45°
d) 90°

42. Given that the vectors $6\mathbf{i} + 15\mathbf{j}$ and $x\mathbf{i} - 5\mathbf{j}$ are parallel, the value of x is

a) -2
b) 2
c) 3
d) -3

43. Given that the matrix

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$$

the order of the transpose of M is

a) 3×2
b) 3×3
c) 2×3
d) 2×2

44. Given that $M = \begin{pmatrix} 3 & -2 \\ 4 & 0 \end{pmatrix}$ and $N = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$, the value of $M + N$ is

a) $\begin{pmatrix} 4 & 1 \\ 6 & 5 \end{pmatrix}$
b) $\begin{pmatrix} 4 & 5 \\ 6 & 5 \end{pmatrix}$
c) $\begin{pmatrix} 4 & -6 \\ 8 & 0 \end{pmatrix}$
d) $\begin{pmatrix} 4 & -5 \\ 8 & 0 \end{pmatrix}$

45. The point $(5, 3)$ when reflected in the line $y = x$ has image

a) $(2, 3)$
b) $(5, 5)$
c) $(5, -3)$
d) $(3, 5)$

46. The range of the distribution $6, 1, 5, 7, 8, 4, 10$ is

a) 4
b) 9
c) 6
d) $5\frac{6}{7}$

47. The modal class in the following frequency distribution table is

Weight	**2 - 5**	**6 - 9**	**10 - 13**	**14 - 17**
frequency	3	5	1	2

a) 14-17
b) 2 - 5
c) 10 - 13
d) 6 - 9

48. The middle quartile of a cumulative frequency curve is at the

a) 75th percentile
b) 25th percentile
c) 50th percentile
d) 100th percentile

49. Let E be an event. Given that $P(E) = \frac{1}{2}$ then $P(E')$ is

Options as printed: A) $\frac{1}{4}$ B) 0.3 C) $\frac{1}{5}$ D) $\frac{2}{3}$

50. Given that 20 students wrote a test out of 10 marks as shown in the following table. The probability of having a student who scored more than 6 marks is

Mark, x	**3**	**4**	**5**	**6**	**7**	**8**
Number of students	2	3	6	4	3	2

a) $\frac{1}{4}$
b) $\frac{3}{20}$
c) $\frac{1}{5}$
d) $\frac{6}{20}$

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The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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